

CUMBERLAND, WI, USA

WMO: 720929

Lat: 45.506N

Lon: 91.981W

Elev: 1241

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 11-19

WBAN: 00316

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-12.7	-24.3	1.5	-15.0	-19.6	2.0	-10.1	24.3	17.2	21.8	18.0	4.8	310	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	84.1	70.7	81.4	68.3	79.2	66.5	73.3	80.9	71.4	78.6	69.3	75.2	10.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	122.4	79.1	68.5	109.6	75.7	67.5	105.9	74.5	37.8	80.9	36.0	79.8	34.1	76.8	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.4	18.9	17.1	DB	-23.8	87.8	5.2	3.0	-27.5	90.0	-30.6	91.8	-33.5	93.5	-37.2	95.7	
			WB	-24.2	75.6	5.1	2.6	-27.9	77.4	-30.8	78.9	-33.7	80.4	-37.4	82.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	13.1	14.7	28.7	40.1	54.4	63.8	67.9	64.7	58.3	44.8	30.5	19.1
	DBStd	21.66	13.89	12.64	12.84	9.38	8.43	5.74	5.27	5.12	8.07	8.38	10.24	12.14
	HDD50	4942	1144	988	670	319	50	0	0	19	210	585	957	
	HDD65	8732	1609	1408	1125	748	342	86	30	70	229	629	1034	1422
	CDD50	1958	0	0	10	21	186	415	554	455	269	47	1	0
	CDD65	274	0	0	0	0	13	51	119	60	29	2	0	0
	CDH74	2252	0	0	1	1	168	421	985	447	211	18	0	0
	CDH80	407	0	0	0	0	34	81	197	61	33	1	0	0
Wind	WSAvg	7.3	8.3	8.1	8.0	8.8	8.0	7.2	5.9	5.0	6.2	7.4	7.9	7.4
Precipitation	PrecAvg	32.4	1.0	0.9	1.7	2.7	3.6	4.2	4.2	3.9	3.7	3.3	1.8	1.1
	PrecMax	43.7	2.3	2.2	3.2	7.4	7.4	6.0	7.2	8.5	7.9	6.9	5.8	2.3
	PrecMin	22.6	0.2	0.1	0.7	0.5	1.3	1.3	0.9	1.2	0.9	1.1	0.2	0.3
	PrecStd	4.9	0.6	0.6	0.7	1.5	1.5	1.4	1.7	1.7	1.9	1.7	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	46.0	68.3	70.5	85.5	86.4	89.9	85.9	83.9	77.0	60.9	44.6
		MCWB	35.6	42.2	60.1	56.1	64.4	71.1	77.3	71.9	69.1	58.5	50.7	41.4
	2%	DB	35.8	38.6	59.2	65.6	79.0	81.7	84.5	81.3	80.6	69.8	54.4	38.9
		MCWB	33.5	34.2	53.1	51.8	62.0	67.0	72.7	69.2	68.0	56.2	47.9	36.2
	5%	DB	33.5	34.4	52.2	61.0	73.1	78.9	81.5	79.0	76.6	63.4	49.8	35.9
		MCWB	32.2	32.0	46.0	48.1	59.8	65.5	69.4	67.0	64.5	54.5	44.5	34.5
	10%	DB	31.8	32.1	45.1	55.2	69.7	76.5	79.3	76.8	72.2	58.7	45.0	32.5
		MCWB	31.0	30.5	40.3	45.2	57.3	63.5	67.3	66.1	63.6	51.0	40.9	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	42.3	60.6	57.1	69.2	74.4	79.2	74.2	72.8	62.3	53.7	42.7
		MCDB	39.3	46.3	68.2	68.4	79.4	83.3	87.0	82.4	79.5	70.1	57.5	43.5
	2%	WB	33.3	35.2	55.2	52.8	64.6	70.8	74.2	72.0	69.8	58.4	48.4	36.8
		MCDB	34.5	37.7	59.0	61.5	72.1	78.6	81.7	79.1	75.0	66.8	54.6	38.3
	5%	WB	32.0	32.3	45.8	50.0	62.5	67.7	72.0	69.6	67.3	55.0	44.3	33.8
		MCDB	32.7	33.6	49.7	58.5	69.6	76.2	79.5	74.2	73.7	62.0	48.5	35.1
	10%	WB	30.3	30.4	40.0	46.5	60.1	65.7	69.9	67.7	65.2	52.4	40.8	32.2
		MCDB	31.1	31.7	44.7	54.7	67.1	73.1	76.4	74.8	70.8	57.3	45.0	32.8
Mean Daily Temperature Range	5% DB	MDBR	15.7	17.4	16.7	17.3	20.4	18.8	19.4	19.6	20.2	16.8	13.6	12.6
		MCDBR	13.6	15.9	22.3	27.0	26.1	22.3	21.0	21.9	23.8	25.2	21.9	13.3
		MCWBR	11.4	13.3	14.3	15.4	12.9	10.9	11.1	11.6	12.7	14.5	14.4	11.5
	5% WB	MCDBR	13.3	14.1	20.4	24.1	21.3	18.9	19.1	18.8	19.0	21.5	19.7	10.8
		MCWBR	12.0	12.2	14.3	15.1	12.4	10.9	11.3	10.9	11.3	13.7	14.1	9.3
Clear-Sky Solar Irradiance	taub		0.252	0.270	0.310	0.352	0.384	0.381	0.372	0.384	0.358	0.313	0.283	0.260
	taud		2.390	2.336	2.356	2.358	2.307	2.346	2.393	2.374	2.424	2.537	2.515	2.425
	Ebn at Noon		278	292	293	288	280	281	282	274	273	274	264	262
	Edh at Noon		23	29	33	36	40	39	36	36	31	24	20	20
All-Sky Solar Radiation	RadAvg		528	877	1233	1459	1688	1865	1956	1635	1270	796	491	364
	RadStd		58	76	110	144	132	143	74	118	83	97	47	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page